

Abstract

A multi-phase research and engineering effort on adapting Large Language Models (LLMs) for efficient edge deployment.

Phase I involves understanding Google Cloud Vertex AI for model training and deployment.

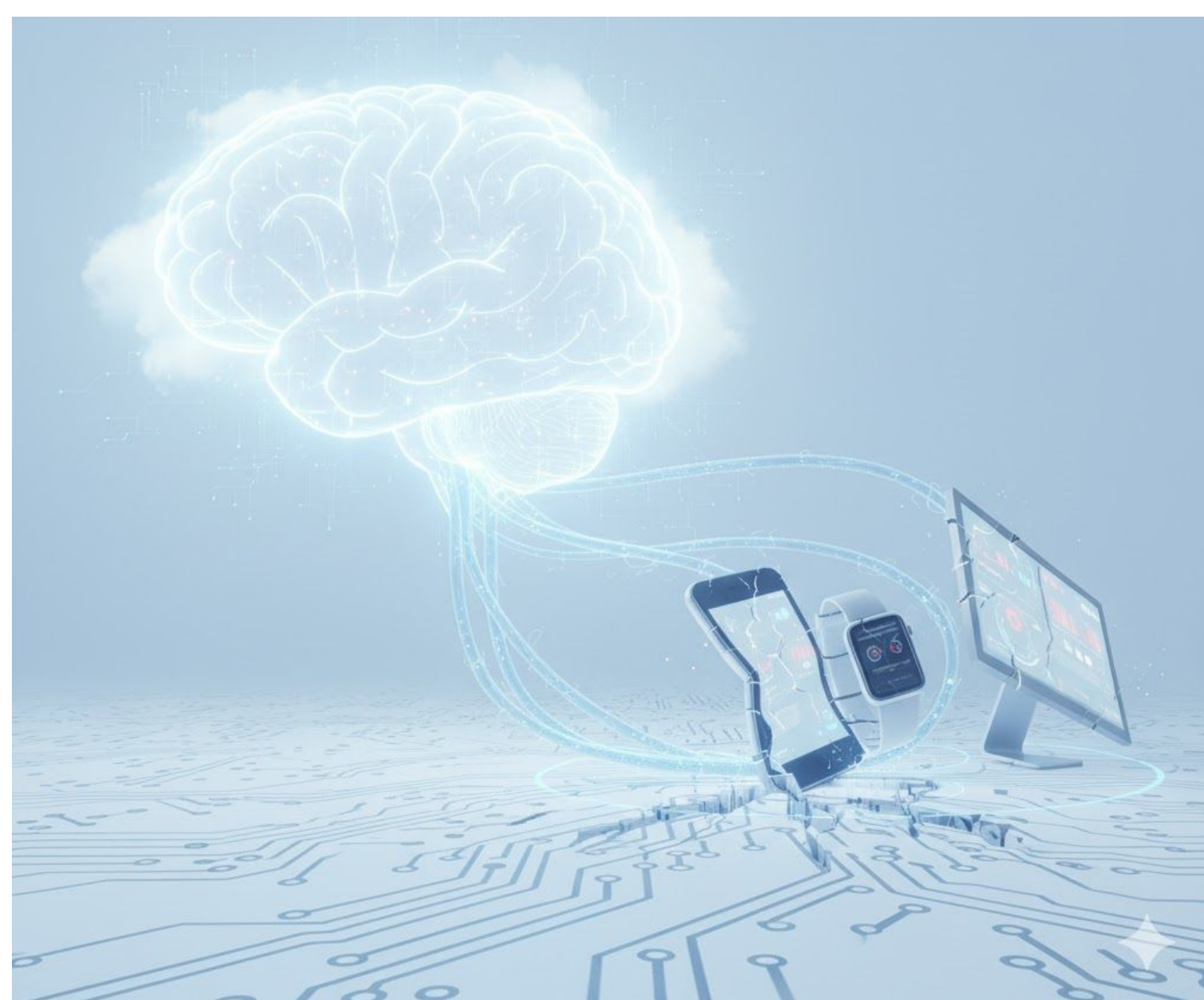
Phase II focuses on compressing open-source LLMs via post-training quantization for embedded systems.

Phase III envisions a decentralized network of compressed LLMs collaborating on edge devices for personalized, privacy-preserving healthcare insights.

Problem / Challenge

- LLMs are computationally intensive and often unsuitable for local deployment.
- Sensitive healthcare data must remain private, limiting reliance on cloud-based inference.
- There is a need for efficient, edge-compatible AI that can process personal data securely and in real time.

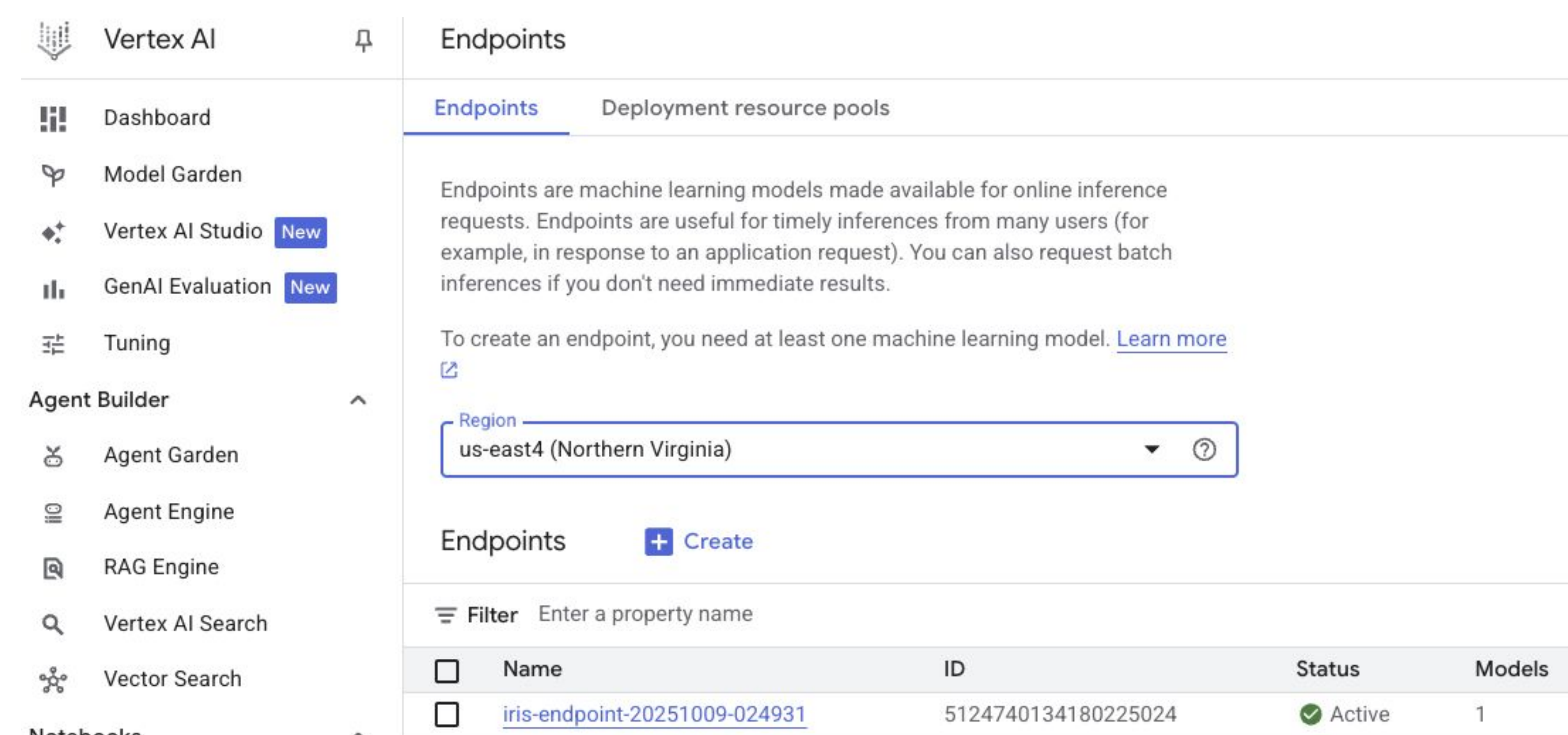
Goal: Develop and evaluate compressed, quantized TinyLLMs capable of operating collaboratively across embedded devices as a blueprint for decentralized health intelligence.



Phase I: Foundations

End-to-End Model Deployment on Google Vertex AI

- **Objective:** Train, register, and deploy a scikit-learn RandomForest model (Iris dataset) to a live Vertex AI endpoint.
- **Key Takeaways:**
 - GCP workflow familiarity
 - Debugging serialization/version mismatches
 - Foundation for distributed model management



Phase II: TinyLLM Compression Project

Evaluating Post-Training Quantization for Edge-Ready LLMs

- **Goal:** Apply GPTQ and AWQ to Phi-3 Mini to enable CPU/embedded inference
- **Method Overview:**
 1. Prepare datasets (WikiText-103, MMLU subset)
 2. Quantize → 4-bit versions
 3. Evaluate latency / perplexity / accuracy / memory footprint
- **Stretch:** Add SparseGPT pruning for further compression.
- **Outcome:** Portable CPU-runnable build (GGUF model) → runs locally via llama.cpp

Phase III (Future Direction): Simulated Edge Network & Healthcare Vision

Edge-RAG for Personalized Health Intelligence

- **Concept:** A network of embedded LLMs (each running a compressed TinyLLM) collaborating through local feedback loops.
- **Simulation Goals:**
 - Emulate multi-device data flows (wearables, mobile app, home monitor)
 - Each node runs quantized LLM with RAG component
 - Periodic knowledge exchange without cloud dependency
- **Healthcare Use Case:**
 - Aggregates signals from multiple edge devices (heart rate, sleep, activity)
 - Synthesizes personalized insights across devices → holistic health view
- **Future Steps:**
 - Implement multi-node simulation in Vertex AI custom jobs or local containers
 - Explore federated learning integration
 - Evaluate privacy vs accuracy trade-offs

References

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